



Predicting Hospital Readmission Risk Using Machine Learning

Capstone Project -1

Abstract

Hospital readmissions increase healthcare costs and reduce operational efficiency. This project predicts patient readmission risk using machine learning

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Business Problem:

Hospital readmissions have become a major concern for healthcare organizations due to their impact on patient outcomes, hospital resource utilization, operational efficiency, and healthcare costs. MediCare Health Network has observed an increasing number of patients being readmitted within a short period after discharge. These readmissions lead to increased treatment costs, reduced bed availability, lower patient satisfaction, and potential financial penalties from insurance providers and regulatory agencies. Current discharge planning primarily relies on manual clinical assessment and standard treatment protocols, making it difficult to consistently identify patients who are at high risk of readmission.

Project Objective:

The primary objective of this project is to develop a machine learning model capable of predicting the likelihood of hospital readmission using historical patient data. The project also aims to:

- Analyse demographic, clinical, laboratory, and hospitalization data in Generalize form.
- Identify key factors influencing hospital readmissions.
- Build and compare predictive classification models.
- Evaluate model performance using standard metrics.
- Provide recommendations to support proactive patient care and discharge planning for Hospital Administration and Operational Team.

Challenges and Constraints:

The project is expected to face several challenges during implementation:

- Missing or incomplete patient records.
- Presence of categorical and numerical variables requiring pre-processing.
- Possible imbalance in readmission classes.
- Identification of the most relevant predictive features.
- Ensuring that the model generalizes well to unseen patient data according to HIPPA and GDPR compliances.
- Maintaining interpretability of the model for healthcare decision-making for Hospital Operational Team.

Project constraints include:

- Use of historical data only.
- No access to real-time Electronic Health Record (EHR) systems.
- Ethical handling of healthcare data and patient confidentiality.

Background Study / Literature Review:

Readmission prediction has become an important application of healthcare analytics. The previous methodologies are being used in Excel form manually to analyse the data. Now days, with the help of Machine Learning Techniques, it become much easier to do such type of analysis. By using Machine Learning, we can improve early identification of high-risk patients compared to traditional rule-based methods.

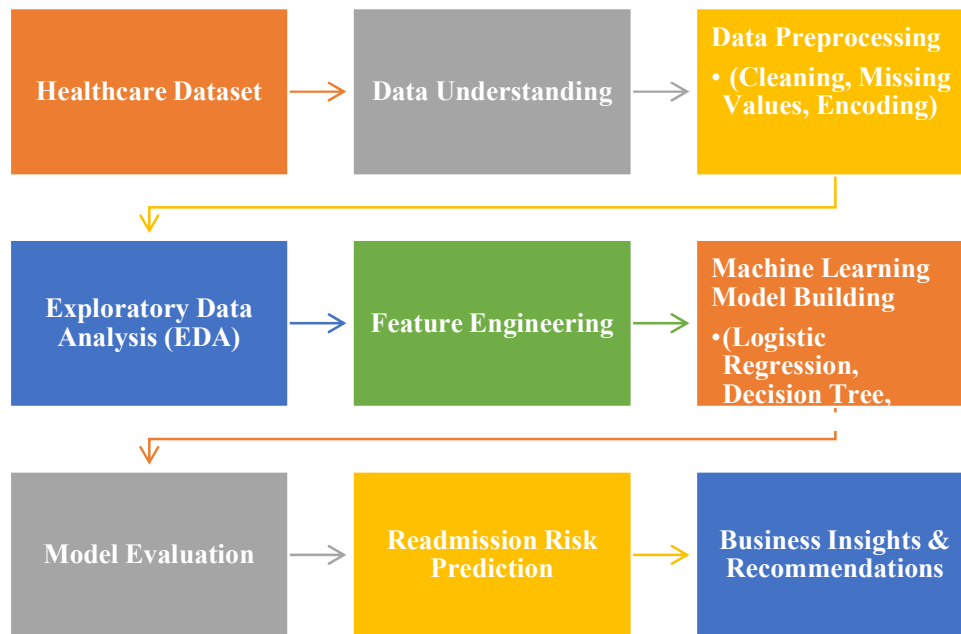
Previous research has shown that factors such as age (**Demographic**), chronic diseases (**History**), previous hospital admissions (**Process**), laboratory results (**Findings**), medication adherence (**Treatment**), and length of hospital stay (**ALOS**) significantly influence readmission risk. Traditional statistical methods such as Logistic Regression remain widely used because of their interpretability, while ensemble learning algorithms such as Random Forest and Gradient Boosting often provide higher predictive accuracy by capturing complex relationships among variables.

Proposed Methodology / Solution Approach:

The project will follow the **CRISP-DM (Cross Industry Standard Process for Data Mining)** framework as per industry standard. Following steps will be follow during analysis:

- A. Business Understanding: Define Project Objectives and success criteria
- B. Data Understanding: Load and explore the dataset. Also, identify data types, missing values and target variable.
- C. Data Preparation: Handle Missing values, encode categorical variables, Scale numerical variables, perform feature engineering.
- D. Exploratory Data Analysis: Need to Analyse patient demographics, correlate clinical variables, visualize correlations and distributions, Identify important predictors.
- E. Develop and compare: Logistic Regression, Decision Tree and Random Forest
- F. Model Evaluation: For Accuracy, Precision, Recall and Confusion Matrix.
- G. Business Recommendations: Need to identify high-risk patients, targeted follow-up strategies, and Improve discharge planning.

Workflow Diagram / Block Diagram:



Expected Deliverables and Output:

The following deliverables are expected upon successful completion of the project:

- **Technical Deliverables:**
 - Cleaned and pre-processed dataset
 - Python Jupyter Notebook
 - Exploratory Data Analysis (EDA)
 - Machine Learning models
 - Model evaluation results
 - Feature importance analysis
- **Documentation:**
 - Capstone Project Report
 - PowerPoint Presentation
 - Source code documentation
- **Business Deliverables:**
 - Patient readmission risk prediction model
 - Identification of high-risk patient groups
 - Recommendations for discharge planning
 - Data-driven insights for reducing avoidable hospital readmissions

Expected Outcome and Output:

The proposed solution is expected to provide an effective predictive framework that assists healthcare professionals in identifying patients with a high probability of readmission before discharge. The project will demonstrate the application of data analytics and machine learning in healthcare while supporting evidence-based decision-making, improving patient care, optimizing hospital resources, and reducing avoidable readmissions.